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Magnetic Resonance Imaging in Prostate Cancer Screening
A Systematic Review and Meta-Analysis

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IMPORTANCE Prostate magnetic resonance imaging (MRI) is increasingly integrated within the prostate cancer (PCa) early detection pathway.

OBJECTIVE To systematically evaluate the existing evidence regarding screening pathways incorporating MRI with targeted biopsy and assess their diagnostic value compared with prostate-specific antigen (PSA)–based screening with systematic biopsy strategies.

DATA SOURCES PubMed/MEDLINE, Embase, Cochrane/Central, Scopus, and Web of Science (through May 2023).

STUDY SELECTION Randomized clinical trials and prospective cohort studies were eligible if they reported data on the diagnostic utility of prostate MRI in the setting of PCa screening.

DATA EXTRACTION Number of screened individuals, biopsy indications, biopsies performed, clinically significant PCa (csPCa) defined as International Society of Urological Pathology (ISUP) grade 2 or higher, and insignificant (ISUP1) PCas detected were extracted.

MAIN OUTCOMES AND MEASURES The primary outcome was csPCa detection rate. Secondary outcomes included clinical insignificant PCa detection rate, biopsy indication rates, and the positive predictive value for the detection of csPCa.

DATA SYNTHESIS The generalized mixed-effect approach with pooled odds ratios (ORs) and random-effect models was used to compare the MRI-based and PSA-only screening strategies. Separate analyses were performed based on the timing of MRI (primary/sequential after a PSA test) and cutoff (Prostate Imaging Reporting and Data System [PI-RADS] score ≥3 or ≥4) for biopsy indication.

RESULTS Data were synthesized from 80 114 men from 12 studies. Compared with standard PSA-based screening, the MRI pathway (sequential screening, PI-RADS score ≥3 cutoff for biopsy) was associated with higher odds of csPCa when tests results were positive (OR, 4.15; 95% CI, 2.93-5.88; $P \leq .001$), decreased odds of biopsies (OR, 0.28; 95% CI, 0.22-0.36; $P \leq .001$), and insignificant cancers detected (OR, 0.34; 95% CI, 0.23-0.49; $P = .002$) without significant differences in the detection of csPCa (OR, 1.02; 95% CI, 0.75-1.37; $P = .86$). Implementing a PI-RADS score of 4 or greater threshold for biopsy selection was associated with a further reduction in the odds of detecting insignificant PCa (OR, 0.23; 95% CI, 0.05-0.97; $P = .048$) and biopsies performed (OR, 0.19; 95% CI, 0.09-0.38; $P = .01$) without differences in csPCa detection (OR, 0.85; 95% CI, 0.49-1.45; $P = .22$).

CONCLUSION AND RELEVANCE The results of this systematic review and meta-analysis suggest that integrating MRI in PCa screening pathways is associated with a reduced number of unnecessary biopsies and overdiagnosis of insignificant PCa while maintaining csPCa detection as compared with PSA-only screening.

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Supplemental content

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Prostate-specific antigen (PSA)–based prostate cancer (PCa) screening has been shown to reduce PCa-specific mortality, but it is also associated with unnecessary biopsies, overdiagnosis, overtreatment, and an unclear effect on overall survival.[1,2] To balance these risks and benefits, clinical practice guidelines recommend shared decision-making strategies to identify informed candidates who are most likely to benefit from PCa early detection.[3,4] However, this opportunistic approach has been associated with widespread but untargeted testing accompanied by disparities in health care access and literacy.[3-5] Moreover, the inherent limitations of PSA-based PCa screening, including excess biopsies and overtreatment of low-grade disease, have not been addressed.[5,6]

Prebiopsy prostate magnetic resonance imaging (MRI) followed by targeted biopsies has been widely integrated in the diagnostic pathway for PCa, as it is associated with improved detection of clinically significant PCa and reduced numbers of avoidable biopsies and insignificant cancers in the clinical setting.[3,7,8] As a result, clinical practice guidelines have recommended prebiopsy MRI; however, to our knowledge, there is no consensus about the role of MRI as an integrated PCa screening tool.[3,4] Consequently, several ongoing clinical trials are investigating the value of incorporating prebiopsy MRI with targeted biopsy into population-based PCa screening protocols to overcome the limitations of conventional PSA-based screening.

In the setting of a large body of literature addressing the diagnostic role of prostate MRI and its growing global use, there is a need to synthesize evidence to inform clinical practice and help devise a screening strategy that incorporates MRI information. To address this unmet need, in this systematic review and meta-analysis, we summarized the currently available literature on the performance of PCa population-based screening strategies that incorporate MRI and compared them with PSA only–based screening approaches. We hypothesized that PSA-MRI–based PCa screening strategies would outperform PSA only–based screening in terms of clinically relevant end points.

Key Points

Question Do prostate cancer screening pathways that incorporate magnetic resonance imaging (MRI) and targeted biopsies outperform strategies that rely solely on prostate-specific antigen testing and systematic biopsy?

Findings In this systematic review and meta-analysis of 80 114 screened men from 12 studies, MRI-based screening was found to be associated with a reduced number of unnecessary prostate biopsies and detection of clinically insignificant prostate cancer while maintaining the detection of clinically significant prostate cancer compared with prostate-specific antigen–only strategies.

Meaning The findings of this meta-analysis support the integration of prostate MRI in prostate cancer screening to improve the balance of patient harms and benefits.

Methods

This systematic review and meta-analysis was reported according to the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline (eTable 1 in Supplement 1) and the Cochrane Handbook.[9,10] The study protocol was registered on PROSPERO.

Eligibility Criteria and Outcome Measures

To evaluate the performance of MRI-based screening strategies, we used the population, intervention, control, and outcomes framework.[11] We included studies of men in the general population or those with elevated genetic risk for PCa who were screened for PCa (population) who underwent MRI examination as part of the screening (intervention) and were compared with men screened for PCa using PSA alone (comparison). Studies were selected if they reported data in screening-like populations, while those addressing diagnostic test accuracy or those that enrolled preselected men to undergo biopsy (men with lower urinary tract symptoms, elevated PSA levels, or suspect digital rectal examination results) were excluded. The primary end point was the cancer detection rate (CDR) of clinically significant PCa, defined as an International Society of Urological Pathology (ISUP) grade of 2 or higher (outcome). Secondary end points included the CDR of insignificant PCa (defined as ISUP grade 1), positive predictive values (PPVs) for detecting significant and insignificant PCas, MRI and biopsy indication, biopsy adherence, and complication rates. Moreover, we calculated CDRs using alternative definitions of significant (ISUP ≥3) and insignificant (ISUP 1-2) PCa. This meta-analysis was restricted to prospective observational or randomized studies.

Search Strategy, Study Selection, and Data Collection

The MEDLINE (via PubMed), Embase, Cochrane/Central, Scopus, and Web of Science databases were queried on May 5, 2023, to identify all available studies containing information on MRI-based screening strategies. After selection by 2 independent review authors (T.F. and G.B.), the following data were extracted from the eligible studies: general information; study population characteristics; details of the intervention and comparator, including screening algorithm (MRI in first-line/sequential screening), sequence (biparametric/multiparametric), and type (1.5T/3T) of MRI; Prostate Imaging Reporting and Data System (PI-RADS) cutoff for the indication of biopsy (PI-RADS ≥3 or ≥4); type of biopsy approach (targeted + systematic/targeted only, cognitive/image-fusion); PSA cutoff; additional novel biomarkers in the screening pathway; and the outcomes of interest described previously.[12] In cases in which studies did not provide information on the specified outcomes, 2 authors (T.F. and G.B.) independently calculated them using the data provided within the studies. Any disagreements on study selection and data extraction were resolved through consensus with a third author (P.R.). Sensitivity, specificity, and negative predictive value could not be evaluated because prostate biopsies were not performed in cases of

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negative screening test results. To address inconsistencies or overlapping data among studies, we made adjustments to the study samples (eTable 2 in Supplement 1). More detailed descriptions of the inclusion criteria, search strategy, selection, and data extraction process are presented in eAppendix 1 and eTables 2 to 3 in Supplement 1.

Statistical Analyses

Quantitative data synthesis was conducted with the R statistical software (version 4.1; R Core Team) and adhered to the methods recommended by the working group of the Cochrane Collaboration.[10] Based on the likely heterogeneity of the studies included, we used random-effect models for our calculations.[13,14] To assess and compare CDR, PPV, MRI, biopsy indication rates, and adherence to biopsy of the different screening pathways, we calculated pooled event rates and odds ratios (ORs) with 95% CIs using the generalized mixed-effect approach.[15] To assess the optimal timing of MRI in the screening pathway, we conducted separate analyses based on different PI-RADS cutoffs for indicating biopsy (≥3, ≥4) and MRI timing (primary/sequential). We used forest plots to visualize event rates and effect measures. To evaluate the moderator effect of different factors, type of MRI sequence, biopsy technique, and study design, we performed subgroup analyses. Heterogeneity was assessed by calculating the I^2 measure and its CI. For all statistical analyses, $P \leq .05$ was considered significant. Publication bias could not be assessed due to the low number of articles for 1 outcome.[16] Full details of the statistical analysis are described in eAppendix 1 in Supplement 1. No ethical approval was required for this systematic review and meta-analysis as already published secondary data were used.

Risk of Bias

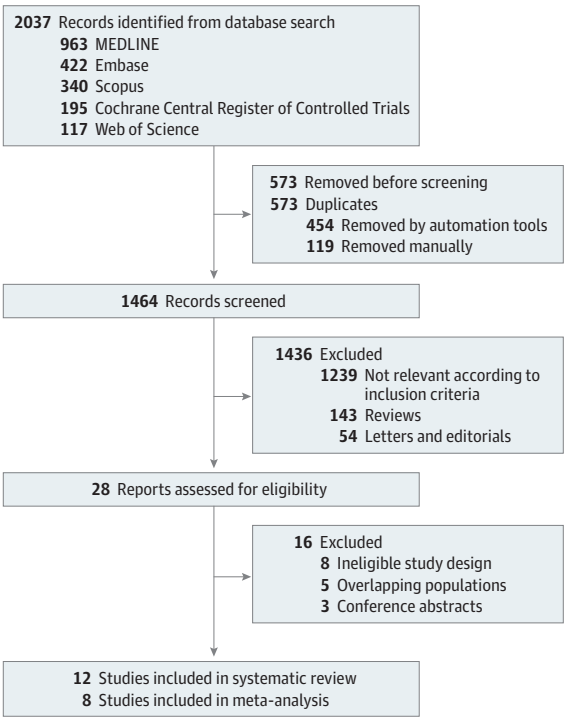
For randomized and nonrandomized studies, the risk of bias (RoB) was evaluated according to the Cochrane Collaboration's RoB assessment (RoB2) and the Risk of Bias in Nonrandomized Studies of Interventions tools independently by 2 reviewers (T.F. and G.B.).[17,18] Disagreements were resolved via consensus with a third author (P.R.).

Results

Study Selection and Baseline Characteristics

The search key identified 2037 studies, of which 1464 were screened after removing duplicates. Finally, from the 28 full-text selected studies, 12 and 8 were eligible for qualitative and quantitative evidence synthesis, respectively (**Figure 1**). A detailed explanation for the exclusion of the studies can be found in eAppendix 2 in Supplement 1. **Table 1**[19-30] shows the baseline characteristics of the included studies. Overall, we assessed 80 114 screened individuals from 12 studies. We identified 4 population-based randomized clinical trials, 2 prospective cohort studies, and 3 prospective pilot studies.[19-25,28-30] Moreover, we included 2 studies that reported on the efficacy of MRI in a prescreened population.[26,27] We identified 4 studies that reported data on the use of novel molecular biomarkers and MRI in PCa screening.[23,28-30] Most publications included data on the use of MRI as a sequential screening tool (after PSA prescreening); however, 3 studies were identified reporting on up-front MRI.[19,24,25] Five studies used biparametric MRI (bpMRI), and 8 included multiparametric MRI (mpMRI) (Table 1). As for the method of biopsy, 7 studies used MRI targeted only, while 6 studies used additional systematic sampling (Table 1). For further details on the studies and interventions, please refer to eTables 4 and 5 in Supplement 1.

Figure 1. Flowchart of the Study Selection Process



MRI as a Sequential Screening Tool

We synthesized data from 57 081 men that were derived from 6 articles that applied MRI in a PSA-prescreened population (as part of sequential screening), with a PI-RADS score of 3 or higher as a cutoff as the biopsy indication.[19-22,26,27] A detailed overview of CDRs, PPVs, biopsy indication, and adherence rates can be found in **Table 2** and eFigures 1 to 4 in Supplement 1. The number of men needed to screen to detect 1 significant PCa was 59 and 63 for PSA only and MRI-based strategies, respectively. Although we found no difference between the MRI-only and PSA-only screening methods in terms of clinically significant CDR (OR, 1.02; 95% CI, 0.75-1.37; $P = .86$) (eFigure 1 in Supplement 1), the MRI pathway was associated with lower odds of insignificant PCa detection (OR, 0.34; 95% CI, 0.23-0.49; $P = .002$) (**Figure 2**).[19,20,22,26,27] These trends in CDR remained similar when alternative definitions were applied for significant (ISUP ≥3: OR, 0.91; 95% CI, 0.54-1.52; $P = .40$) and insignificant PCas (ISUP 1-2: OR, 0.54; 95% CI, 0.23-1.29; $P = .09$) (eFigure 5 in Supplement 1). Furthermore, screening strategies that incorporated MRI had a higher

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Table 1. Characteristics of the Included Studies

Study (study name)	Year	Country	Study design	Age of screened men, median (IQR), y	No. screened	Details of MRI-based strategy (intervention)			Details of PSA-based strategy (comparator)	
						Indication of MRI	MRI type	Method of biopsy	PSA cutoff	Type of biopsy
Eldred-Evans et al[19] (IP1-PROSTAGRAM)	2023	UK	Prospective cohort	57 (53-61)	All: 408[a]	First-line screening and PSA ≥3 ng/mL	bpMRI	Image fusion transperineal targeted	≥3 ng/mL	Transperineal systematic
Hugosson et al[20] (Göteborg-2, first round)	2022	Sweden	RCT	56 (52-59)	Intervention: 11 986; comparator: 5994	PSA, ≥3 ng/mL	mpMRI	Cognitive transrectal targeted[b]	≥3 ng/mL	Transrectal systematic[c]
Arsov et al[21] (PROBASE)	2022	Germany	RCT	Mean (range), 45 (44-47)	Intervention: 23 341; comparator: 23 301	PSA ≥3 ng/mL	mpMRI[d]	Image fusion transrectal targeted and systematic	NA	NA
Eklund et al[22] (STHLM3-MRI)	2021	Sweden	RCT	66 (61-71)	Intervention: 929[e]; comparator: 603[e]	PSA ≥3 ng/mL	bpMRI	Image fusion transrectal targeted and systematic[f]	≥3 ng/mL	Transrectal systematic
Nordström et al[23] (STHLM3-MRI)	2021	Sweden	RCT	66 (61-71)	Intervention: 1372[g]; comparator: 921[g]	PSA ≥3 ng/mL or Stockholm 3 score ≥0.11	bpMRI	Image fusion transrectal targeted and systematic	≥3 ng/mL	Transrectal systematic
Nam et al[24] (MVP; pilot study)	2016	Canada	Prospective cohort	61 (55-68)	All: 47	First-line screening	mpMRI	Cognitive transrectal targeted and systematic	NA	NA
Nam et al[25] (MVP)	2022	Canada	RCT	Mean (SD), 68 (7.3)	Intervention: 259; comparator: 266	First-line screening	bpMRI	Image fusion transrectal targeted and systematic	≥2.6 ng/mL	Transrectal systematic
Grenabo Bergdahl et al[26] (Göteborg, tenth round; pilot study)	2016	Sweden	Prospective cohort	69 (69-70)	All: 384	PSA ≥3 ng/mL	mpMRI	Cognitive transrectal targeted	≥3 ng/mL	Transrectal systematic
Alberts et al[27] (ERSPC, fifth round; pilot study)	2018	Netherlands	Prospective cohort	73 (72-73)	All (arm 2): 713	PSA ≥3 ng/mL	mpMRI	Image fusion transrectal targeted	≥3 ng/mL	Transrectal systematic
Rannikko et al[28] (ProScreen; pilot study)	2022	Finland	Prospective cohort	64-65	All: 170	PSA ≥3 ng/mL and 4Kscore >7.5%	mpMRI	Image fusion transrectal targeted	NA	NA
Benafif et al[29] (BARCODE-1; pilot study)	2022	UK	Prospective cohort	Mean (range), 61 (55-69)	All: 307	Poligenic risk score ≥90th percentile	mpMRI	Image fusion transrectal targeted and systematic[h]	NA	NA
Segal et al[30] (NCT02053805)	2020	Israel	Prospective cohort	Mean (SD), 54 (9.8)[i]	*BRCA1*: 108; *BRCA2*: 80	Elevated age-stratified PSA level[j]	mpMRI	Image fusion transrectal targeted	Elevated age-stratified PSA level[j]	Transrectal systematic

Abbreviations: 4Kscore, kallikrein panel; bpMRI, biparametric MRI; mpMRI, multiparametric MRI; MRI, magnetic resonance imaging; NA, not applicable; PI-RADS, Prostate Imaging–Reporting and Data System; RCT, randomized clinical trial; PSA, prostate-specific antigen; PSAD, PSA density.

SI conversion factor: to convert PSA to µg/L, multiply by 1.

[a] Patients underwent screening with PSA and MRI; therefore, MRI as a first- and second-line (after PSA) screening tool was assessed.

[b] For negative MRI and PSA ≥10 ng/mL, systematic biopsy was performed. To assess the performance of targeted biopsy only, we excluded cancers detected with systematic biopsy and negative MRI in the experimental arm.

[c] For positive MRI in the reference arm, targeted and systematic biopsy were performed. To assess performance of systematic biopsy only, we excluded cancers detected with targeted biopsy in the reference arm.

[d] MRI examination was not part of PROBASE screening since the trial was started before mpMRI was recommended for primary diagnosis of prostate cancer. However, data on MRI were available for 79% of participants, and 95% underwent MRI/ultrasonography fusion-targeted and systematic biopsy. Arm 2 of this study indicated prostate biopsy solely on rectal digital examination findings and was not included.

[e] Numbers represent patients with a PSA ≥3 ng/mL, as randomization was performed after PSA prescreening. Initially, 12 750 patients were screened with PSA.

[f] For negative MRI and Stockholm 3 score ≥0.25, systematic biopsy was performed. To assess the performance of MRI-based biopsy only, we excluded cancers detected with systematic biopsy based on an elevated Stockholm 3 test.

[g] Numbers represent patients with a PSA ≥3 ng/mL or a Stockholm 3 score ≥0.11, as randomization was performed after PSA and Stockholm 3 score-based prescreening. Initially, 12 750 patients were screened.

[h] All patients with a polygenic risk score ≥90th percentile underwent MRI and systematic biopsy. For positive MRI results (PI-RADS score ≥3), targeted biopsy was added.

[i] This study enrolled patients with pathogenic germline breast cancer gene 1 or 2 alterations.

[j] Elevated age-stratified PSA was defined as ≥1 ng/mL for those aged 40 to 50 years, ≥2 ng/mL for those aged 50 to 60 years, and ≥2.5 ng/mL for those aged 60 to 70 years.

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Table 2. Diagnostic Performance of Screening Strategies Incorporating MRI[a]

Screening type	Cancer detection rate (95% CI), %		Positive predictive value (95% CI), %		Biopsy indication rate (95% CI), %	Biopsy adherence rate (95% CI), %
	Significant PCa	Insignificant PCa	Significant PCa	Insignificant PCa		
MRI sequential (PI-RADS 3-5)						
MRI	1.1 (0.4-3.1); I^2: 98%	0.4 (0.1-1.4); I^2: 94%	41.9 (28.5-56.7); I^2: 90%	16.3 (10.8-23.9); I^2: 67%	2.9 (1.4-6.2); I^2: 99%	95.9 (77.1-99.4); I^2: 95%
PSA	1.7 (1.0-2.8); I^2: 86	1.9 (0.7-4.6); I^2: 96%	16.1 (10.4-24.2); I^2: 76%	18.4 (11.9-27.3); I^2: 74%	13.2 (7.3-22.8); I^2: 98%	88 (75.1-94.6); I^2: 93%
MRI vs PSA (OR)	1.02 (0.75-1.37)	0.34 (0.23-0.49)	4.15 (2.93-5.88)	1.0 (0.5-2.0)	0.28 (0.22-0.36)	4.61 (2.39-8.89)
P value	.86	.002	.001	.99	.001	.01
MRI sequential (PI-RADS 4-5)						
MRI	1.2 (0.4-3.9); I^2: 86%	0.4 (0.2-0.7); I^2: 45%	48.9 (35.4-62.6); I^2: 0%	21.1 (11.9-34.7); I^2: 0%	2.4 (0.9-6.3); I^2: 89%	98.7 (86.6-99.9) I^2: 0%
PSA	1.4 (0.4-4.7); I^2: 87%	1.9 (0.2-17.5); I^2: 98%	14.9 (9.5-22.7); I^2: 15%	20.9 (7.7-45.5); I^2: 74%	11.9 (2.9-38.2); I^2: 99%	90.5 (72.2-97.2); I^2: 76%
MRI vs PSA (OR)	0.85 (0.49-1.45)	0.23 (0.05-0.97)	7.01 (1.76-27.98)	0.99 (0.29-3.32)	0.19 (0.09-0.38)	4.68 (0.37-59.49)
P value	.23	.048	.03	.96	.01	.12
MRI primary (PI-RADS 4-5)						
MRI	6.0 (0.6-39.4); I^2: 92%	1.2 (0.2-7.3); I^2: 55%	41.9 (16.1-73.0); I^2: 57%	10.1 (2.2-35.9); I^2: 0%	15.0 (3.1-49.7);I^2: 91%	93.1 (48.1-99.5); I^2: 0%
PSA	NA	NA	NA	NA	18.1 (4.7-49.7); I^2: 91%	NA
MRI vs PSA (OR)	NA	NA	NA	NA	0.81 (0.23-2.87)	NA
P value	NA	NA	NA	NA	.53	NA

Abbreviations: MRI, magnetic resonance imaging; NA, not applicable; OR, odds ratio; PCa, prostate cancer; PI-RADS, Prostate Imaging–Reporting and Data System; PSA, prostate-specific antigen.

[a] We evaluated MRI as primary or sequential screening tool and PI-RADS cutoffs of 3 or 4 for the biopsy indication. Pooled rates are represented in percentages with 95% CIs. Between-study heterogeneity is expressed by I^2 values. For the comparison of MRI-based and PSA-based screening, we calculated ORs with 95% CIs.

PPV for detecting significant PCa (OR, 4.15; 95% CI, 2.93-5.88; *P* = .001) and a lower biopsy rate (OR, 0.28; 95% CI, 0.22-0.36; *P* < .001) than PSA-only-based ones (Figure 2; eFigures 2 and 3 in Supplement 1).[19,20,22,26,27] The pooled rate of MRI was 8.5% (95% CI, 2.6%-24.8%; I^2 = 100%) among the screened individuals, and biopsy adherence was higher when MRI was used (OR, 4.61; 95% CI, 2.39-8.89; *P* = .01) (eFigure 4 in Supplement 1).[19-22,26,27] To identify the high rate of heterogeneity among the studies and assess the role of possible confounders, we stratified studies based on the type of MRI sequence, biopsy method, and study design (eFigures 6 and 7 in Supplement 1). We observed differences in terms of PPV, but not in CDR and biopsy rates. Compared with mpMRI, the use of bpMRI was associated with a higher PPV for significant PCa (61.1% [95% CI, 26.5%-87.3%] vs 34.8% [95% CI, 25.2%-45.7%]; *P* < .001) and a lower PPV for insignificant PCa (11.5% [95% CI, 1.3%-55.1%] vs 19.5% [95% CI, 12.3%-29.6%]; *P* = .01), respectively, without heterogeneity across the subgroups (eFigures 7A and 7E in Supplement 1) .Targeted + systematic (vs targeted) and image fusion (vs cognitive) biopsies had a lower PPV for insignificant cancers (eFigures 7F and 7G in Supplement 1).

Among 19 501 patients who underwent prostate MRI that used a PI-RADS cutoff of 4 or higher as a biopsy indication, we observed even lower odds of insignificant PCa detection (OR, 0.23; 95% CI, 0.05-0.97; *P* = .048) and biopsy (OR, 0.19; 95% CI, 0.09-0.38; *P* = .01) with a higher PPV (OR, 7.01; 95% CI, 1.76-27.98; *P* = .03) and similar CDR (OR, 0.85; 95% CI, 0.49-1.45; *P* = .23) for significant disease compared with standard PSA-only screening (**Figure 3**; eFigures 8-10 in Supplement 1).[19,20,27]

MRI as a First-Line Screening Tool

To evaluate the performance of MRI (PI-RADS ≥4) as a primary screening tool, we synthesized data from 3 articles that involved 983 men.[19,24,25] Clinically significant and insignificant CDRs were 6% (95% CI, 0.6%-39.4%; I^2: 92%) and 1.2% (95% CI, 0.2%-7.3%; I^2: 55%), respectively (eFigures 11A and 11B in Supplement 1). The PPV of up-front MRI to detect significant PCa was 41.9% (95% CI, 16.1%-73%; I^2: 57%) (eFigure 11C in Supplement 1). Due to the limited availability of data, comparison of MRI-based screening with PSA-based approaches was only feasible in terms of biopsy selection, which revealed no significant difference between the 2 strategies (OR, 0.81; 95% CI, 0.23-2.87; *P* = .50) (eFigure 12C in Supplement 1).[19,24,25]

MRI-Based and Biomarker-Based Screening Strategies

We identified 4 articles that reported on the combination of MRI and novel biomarkers; however, given the heterogeneity

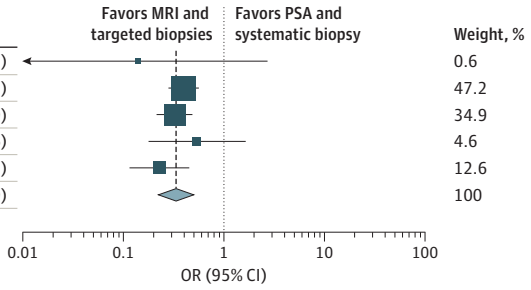
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Figure 2. Comparison of Magnetic Resonance Imaging (MRI) and Standard Prostate-Specific Antigen (PSA)–Based Screening Strategies in Terms of Prostate Cancer Detection, Biopsy Indication Rates, and Positive Predictive Value (PPV)

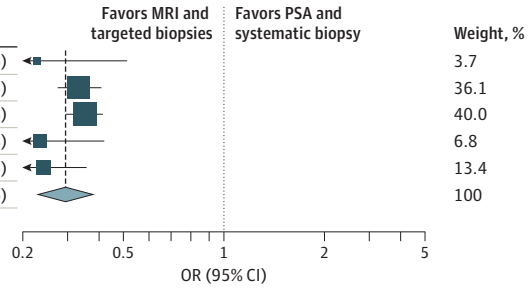
A CDR of clinically insignificant prostate cancer

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	0	408	3	408	0.14 (0.01-2.75)	0.6
Hugosson et al,[20] 2022	59	11 986	73	5994	0.40 (0.28-0.57)	47.2
Eklund et al,[22] 2021	36	7650	73	5100	0.33 (0.22-0.49)	34.9
Grenabo Bergdahl et al,[26] 2016	5	384	9	384	0.55 (0.18-1.66)	4.6
Alberts et al,[27] 2018	11	713	45	713	0.23 (0.12-0.45)	12.6
Random-effects model		21 141		12 599	0.34 (0.23-0.49)	100
Test overall: $P = .002$						



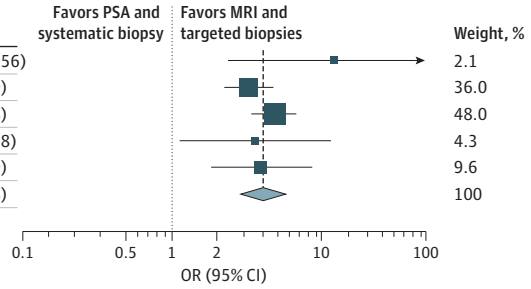
B Biopsy indication rates

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	10	408	41	408	0.22 (0.11-0.46)	3.7
Hugosson et al,[20] 2022	268	11 986	405	5994	0.32 (0.27-0.37)	36.1
Eklund et al,[22] 2021	325	7650	603	5100	0.33 (0.29-0.38)	40.0
Grenabo Bergdahl et al,[26] 2016	21	384	77	384	0.23 (0.14-0.38)	6.8
Alberts et al,[27] 2018	48	713	167	713	0.24 (0.17-0.33)	13.4
Random-effects model		21 141		12 599	0.28 (0.22-0.36)	100
Test overall: $P < .001$						



C PPV for clinically significant prostate cancer

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	6	10	4	37	12.38 (2.41-63.56)	2.1
Hugosson et al,[20] 2022	104	262	58	348	3.29 (2.26-4.79)	36.0
Eklund et al,[22] 2021	186	304	106	438	4.94 (3.59-6.78)	48.0
Grenabo Bergdahl et al,[26] 2016	7	20	9	70	3.65 (1.15-11.58)	4.3
Alberts et al,[27] 2018	17	48	19	158	4.01 (1.87-8.59)	9.6
Random-effects model		644		1051	4.15 (2.93-5.88)	100
Test overall: $P = .001$						



Screening pathways incorporating MRI were associated with a reduced number of clinically insignificant disease (A) and biopsies (B); moreover, they also outperformed PSA-only strategies in terms of PPV for clinically significant prostate cancer detection (C). MRI was applied as a sequential screening tool (after PSA-prescreen), with a Prostate Imaging Reporting and Data System score of 3 or higher as a cutoff for biopsy indication. CDR indicates cancer detection rate; OR, odds ratio.

between populations and interventions within studies, we did not perform a quantitative data synthesis.[23,28-30] In this subset, the use of novel biomarkers was associated with fewer insignificant PCas while maintaining significant disease detection.[23,28] Moreover, MRI has been shown to be an effective screening tool in patients with a genetic predisposition for PCa.[30]

RoB

The RoB 2 and Risk of Bias in Nonrandomized Studies of Interventions tools identified a low overall RoB in most of the included studies for the CDR, PPV, MRI, and biopsy rates and adherence to biopsy indication outcomes (eFigures 13 and 14 in Supplement 1). Among randomized clinical trials, the intervention in the PROBASE trial was found to be biased, as MRI examination was not part of the screening protocol; however, MRI data were available in 79% of participants, and 114 of 120 men (95%) underwent MRI/ultrasonography fusion-targeted and systematic biopsy.[21] Despite some prospective cohort studies that showed a moderate risk in categories that was mainly based on the population of the study, most of these articles displayed a low overall RoB.

Discussion

To our knowledge, this systematic review and meta-analysis was the first to assess the performance of MRI in the setting of PCa screening. There were several notable and clinically relevant findings from this study. First, these analyses suggested that MRI as part of sequential screening performs similarly to conventional PSA-based strategies in detecting clinically

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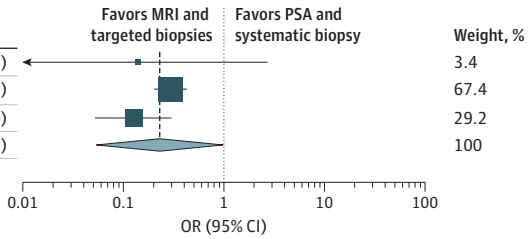
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Figure 3. Performance of Magnetic Resonance Imaging (MRI) With a Prostate Imaging Reporting and Data System (PI-RADS) Score of 4 or Higher as a Cutoff for Biopsy Indication

A CDR of clinically insignificant prostate cancer

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	0	408	3	408	0.14 (0.01-2.75)	3.4
Hugosson et al,[20] 2022	44	11 986	73	5994	0.30 (0.21-0.43)	67.4
Alberts et al,[27] 2018	6	713	45	713	0.13 (0.05-0.30)	29.2
Random-effects model		13 107		7115	0.23 (0.05-0.97)	100

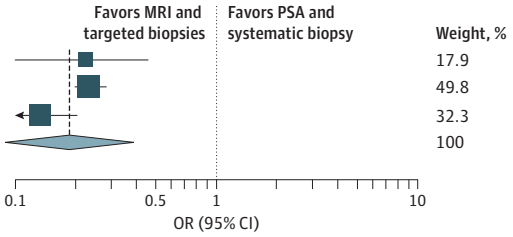
Test overall: *P* = .048



B Biopsy indication rates

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	10	408	41	408	0.22 (0.11-0.46)	17.9
Hugosson et al,[20] 2022	202	11 986	405	5994	0.24 (0.20-0.28)	49.8
Alberts et al,[27] 2018	28	713	167	713	0.13 (0.09-0.20)	32.3
Random-effects model		13 107		7115	0.19 (0.09-0.38)	100

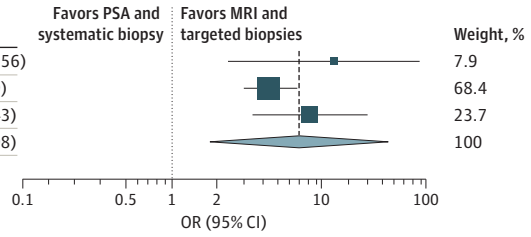
Test overall: *P* = .01



C PPV for clinically significant prostate cancer

Source	Events by MRI, No.	Total MRIs, No.	Events by PSA, No.	Total PSA tests, No.	OR (95% CI)	Weight, %
Eldred-Evans et al,[19] 2023	6	10	4	37	12.38 (2.41-63.56)	7.9
Hugosson et al,[20] 2022	95	199	58	348	4.57 (3.07-6.79)	68.4
Alberts et al,[27] 2018	15	28	19	158	8.44 (3.49-20.43)	23.7
Random-effects model		237		543	7.01 (1.76-27.98)	100

Test overall: *P* = .03



Elevating the PI-RADS cutoff to 4 for the indication of biopsy was associated with lower biopsy rates (B) and less insignificant prostate cancers found (A) compared with prostate-specific antigen (PSA). Moreover, it enhanced the positive predictive value (PPV) for significant disease detection (C). MRI was applied as a sequential screening tool (after PSA prescreening) in this analysis. CDR indicates cancer detection rate; OR, odds ratio.

significant PCa while reducing the number of detected insignificant cancers. Second, prebiopsy MRI was associated with a substantially reduced number of unnecessary prostate biopsies performed and enhanced the PPV for significant PCa detection compared with PSA-only screening with standard biopsies. Moreover, modifying the threshold of offering prostate biopsy to a PI-RADS score of 4 or higher and the use of bpMRI may further be associated with a reduced rate of unnecessary biopsies while not meaningfully compromising the detection of significant PCa. Finally, the results of this study suggest that MRI as a first-line screening tool does not seem to exhibit the aforementioned benefits in reducing biopsy rates and the detection of insignificant PCa.

Our findings support and potentially strengthen the cumulative evidence suggesting that the use of MRI following initial PSA prescreen is associated with decreased detection of insignificant PCa compared with PSA-only approaches. Thus, MRI may be a useful tool to mitigate the limitations of PSA-based screening, including overdiagnosis of indolent PCa, which can be associated with overtreatment with avoidable complications associated with any therapy.[31,32] However, the 2 screening strategies were similar in terms of CDR for clinically significant disease. Based on our analysis, the number needed to screen to detect 1 significant PCa was 59 and 63 for PSA-only and MRI-based strategies, respectively.

Moreover, use of MRI-based screening strategies was associated with higher PPV for detecting clinically significant PCa and a reduced number of biopsy indications. Based on our findings, the number of biopsies needed to detect 1 significant prostate cancer was 2 and 6 with MRI-based and PSA-only screening, respectively. These findings are notable given the risks of bleeding, infection, discomfort, and expense associated with prostate biopsy, as well as the psychological burden of screening-triggered workup.[33,34] Moreover, avoiding biopsy and following up with patients with negative MRI results were shown to be a safe approach in screening.[35,36] According to the data presented in this article, patients are more willing to undergo biopsy when the indication is underlined with MRI results, which is an important factor in achieving better outcomes and a more equal distribution of health care resources.[5,37-39] In modeling studies, MRI-based PCa screening is associated with an improvement in the benefit-harm profile, quality of life, cost-

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effectiveness, and environmental effect of screening for PCa compared with standard PSA-based screening.[40-44] Accordingly, our results synthesizing high-quality prospective data suggested that MRI is effective at identifying individuals who are most likely to require further evaluation and biopsy, potentially reducing the burden on health care resources and sparing patients from having to undergo unnecessary invasive procedures.

This study aggregates performance characteristics of MRI-based screening across PI-RADS cutoffs for biopsy selection, different sequences (multiparametric or biparametric), biopsy methods (targeted only or targeted + systematic), and fusion types (cognitive or image fusion). Our analysis suggests that implementing a PI-RADS score of 4 or higher as a cutoff is further associated with a reduced number of insignificant cancers detected and biopsies performed. Additionally, the choice of MRI sequence, whether biparametric or multiparametric, is an important aspect of screening, as shorter bpMRI protocols are faster, more cost-effective, and are associated with reduced exposure to contrast material, making them valuable in the screening process.[45,46] However, bpMRI interpretation may be more challenging, requiring a higher level of expertise.[47] We found that bpMRI is associated with a higher PPV for detecting significant PCa, which may be attributable to preferentially identifying larger, more conspicuous lesions in the absence of contrast.[45,48] Lastly, we examined the role of biopsy approach on MRI-based screening outcomes. These results revealed no significant differences in terms of CDR and PPV for significant disease between the targeted-only and targeted + systematic biopsy techniques, as well as between image fusion and cognitive biopsy methods. However, the targeted + systematic and image-fusion biopsies demonstrated a lower PPV for detecting clinically insignificant prostate cancer. These findings suggest that a screening pathway incorporating bpMRI following PSA prescreening coupled with a PI-RADS score of 4 or higher as a cutoff for biopsy indication may be a promising strategy for increasingly accessible and cost-effective screening. However, several questions remain to be addressed in future investigations, including whether to use targeted-only biopsy, the optimal method for fusion biopsy, a comprehensive analysis of screening costs, and an examination of long-term survival outcomes. Furthermore, differences in oncologic risk profiles have been observed between PCa cases diagnosed via MRI-based targeted biopsy and those identified through standard biopsy methods.[49] These findings underscore the need for further research to elucidate the behavior of PCas identified with MRI and targeted biopsy and their implications for treatment strategies.

Our study also highlights the importance of considering the timing and type of MRI and biopsy in the screening process. While MRI following PSA prescreening (sequential pathway) demonstrated numerous advantages compared with PSA-only strategies, up-front MRI as a primary tool did not appear to exhibit the aforementioned benefits in terms of biopsy rates and insignificant PCa detection; however, it was associated with CDR for significant PCa. Although these results are limited by the lack of data for formal statistical comparison, this suggests that while MRI is valuable for refining the selection of patients for biopsy, its use as a primary screening tool needs to be further assessed in the future. Interestingly, among men younger than 55 who harbor breast cancer gene (*BRCA*) germline alterations, up-front MRI has been demonstrated to have the highest clinical benefit, highlighting its diagnostic value for patients with a genetic predisposition for PCa.[30]

Limitations

This study had several limitations. These included a relatively low number of articles that could be included. Subgroup evaluation, heterogeneity, and publication bias assessment were limited. As no biopsy was performed in case of a negative MRI result, sensitivity, specificity, and negative predictive values could not be assessed. Most of the studies assessed a Scandinavian population, limiting the generalizability of our findings. Safety and long-term survival data could not be synthesized, constraining the full-scale interpretation of our results. Finally, the optimal intensity and interval of MRI-based screening rounds have yet to be established, which require consideration of trade-offs regarding frequency of procedures, cancer detection, and associated costs.

Conclusions

The results of this systematic review and meta-analysis suggest that prostate MRI with targeted biopsies is an effective strategy for the early detection of prostate cancer. We found that MRI mitigates pitfalls of standard PSA-based strategies, as it can be associated with fewer unnecessary biopsies and helps to avoid the detection of insignificant cancers while not compromising significant disease detection. Our results highlight the need to reassess our approach to population-based screening; however, the optimal setup of MRI and biopsy scheme in the screening process requires further evaluation.

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Other—validation, software, data curation, methodology: Shim.

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